

Not just a reporting tool.

The emissions calculation engine built around your operations.

NeoCarbon connects operational activity data to engineering-grade greenhouse-gas calculations, speciated methane accounting, reporting and verifier-ready decision support across the upstream, midstream and downstream value chain.

One calculation architecture

Tier 1 • Tier 2 • Tier 3

Use default factors, regional references or engineering-based calculations according to available operational data fidelity.

Source-level detail

Well → Facility → Source

Keep the relationship between input data, method, factor, calculation and reported result fully visible and auditable.

WHAT MAKES NEOCARBON DIFFERENT: 6 FUNCTIONAL MODULES

01 / CALCULATION

Engineering-based formulas

API Compendium 2021 algorithms, flash EOS, and physical properties drive process calculations instead of generic factors.

02 / REPORTING

Comprehensive multi-scope

Scopes 1, 2 and 15 Scope 3 tiers. 1-click verified export packages for ISO 14064, EU CBAM, and corporate CSRD formats.

03 / SCENARIOS

Scenario & planning tools

What-if simulation, Marginal Abatement Cost Curves (MACC), flare gas monetization payback, and VRU capital returns.

04 / METHANE

OGMP 2.0 Gold workflow

Component-level source tracking, bottom-up LDAR reconciliation with drone/satellite scans, and continuous reduction roadmaps.

05 / COMPOSITION

Custom fuel & gas composition

Represent real gas chromatography (C1-C6+, CO₂, N₂, H₂S) and calculate constituent-specific mass emissions where data exists.

06 / CONTROL

Audit history & air-gapped security

Full lineage traceability with SHA-256 audit seals. 100% on-premise air-gapped deployment option behind your firewall.

The Calculation Pipeline (Raw Data → Decision Intelligence)

Input → Conversion → Tier / Method → Factor / Formula → CO₂ / CH₄ / N₂O → CO₂e

- Scopes 1, 2 and 3 architecture across all operational entities.
- Carbon, methane and flaring intensity metrics (tCO₂e/boe).
- Direct SCADA/historian telemetry & automated ingestion connectors.
- Dashboard, Excel and PDF reporting ready for DNV/KPMG audit.

Built Around Oil & Gas Emission Sources

Stationary combustion
Boilers Turbines Heaters Generators
Fuel quantity, heating value and stoichiometric combustion factors.

Fugitive & LDAR
Valves Pumps Compressors Flanges
Component-level inventories and methane screening workflows.

Flaring & venting
Routine Non-routine Gas Flaring
Unloading
Destruction efficiency (DRE), volume and gas composition modelling.

Complete value chain
Production Processing Pipelines
Refining
Organize assets, joint ventures and facilities across operations.

HOW NEOCARBON COMPARES

FEATURE / CAPABILITY	NeoCarbon	Other Industry Software	Specialized Methane Tools	Spreadsheets
Full oil & gas coverage (upstream → downstream)	✓ Comprehensive	Partial	Partial	X
Engineering formulas & custom factors (Tier 3)	✓ Full Physics	Partial	X	X
OGMP 2.0 ready (Tier 1/2/3 & Gold Standard)	✓ Built-in Gold L5	Partial	Partial	X
Regulatory reporting (ISO 14064, API, EU, CBAM)	✓ 1-Click Dossiers	Partial	X	X
Data sovereignty & 100% on-premise air-gap	✓ Air-Gapped Behind Firewall	X (Cloud only)	Limited	Unencrypted

REGULATORY STANDARDS & VERIFICATION READINESS

ISO 14064-1 • 14064-2

OGMP 2.0 Gold Standard

API Compendium 2021

EU Methane Reg 2024/1787

EU CBAM

GHG Protocol

Deterministic engineering engines provide full audit lineage, shielding against punitive cross-border EU default tax markups and reducing audit durations by 80%.

Deploy Sovereign Carbon Accounting in 15 Minutes

Provide one gas lab analysis sheet or monthly production report. Our team will model your facility live, calculate your exact Methane Intensity, and show your full profile on the spot.

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